

Formula Finder

Formulas

This document contains formulas and mathematical expressions.

$$v = \frac{d}{t} \quad \text{velocity } [v] = \frac{m}{s}, \text{ distance } [d] = m, \text{ time } [t] = s$$

$$\rho = \frac{m}{V} \quad \text{density } [\rho] = \frac{kg}{m^3}, \text{ mass } [m] = kg, \text{ volume } [V] = m^3$$

$$p = \frac{F}{A} \quad \text{pressure } [p] = \frac{N}{m^2}, \text{ force } [F] = N, \text{ area } [A] = m^2$$

$$W = F \cdot \Delta s \quad \text{work } [W] = J, \text{ force } [F] = N, \text{ displacement } [\Delta s] = m$$

$$P = \frac{E}{t} \quad \text{power } [P] = W, \text{ energy } [E] = J, \text{ time } [t] = s$$

$$\eta = \frac{E_{out}}{E_{in}} \quad \text{efficiency } [\eta] = 0 - 1, \text{ output energy } [E_{out}] = J, \text{ input energy } [E_{in}] = J$$

$$\Delta E = Q + A \quad \text{change in energy } [\Delta E] = J, \text{ heat } [Q] = J, \text{ work done } [A] = J$$

$$C = c \cdot m \quad \text{heat capacity } [C] = J/K, \text{ specific heat capacity } [c] = J/(kg \cdot K), \text{ mass } [m] = kg$$

$$C_{system} = c_A m_A + c_B m_B \quad \text{system heat capacity } [C_{system}] = J/K, \text{ specific heat capacity } [c_A, c_B] = J/(kg \cdot K), \text{ mass}$$

$$Q = C \Delta t \quad \text{heat } [Q] = J, \text{ heat capacity } [C] = J/K, \text{ change in temperature } [\Delta t] = K$$

$$Q = cm \Delta t \quad \text{heat } [Q] = J, \text{ specific heat capacity } [c] = J/(kg \cdot K), \text{ mass } [m] = kg, \text{ change in temperature } [\Delta t] = K$$

$$Q = L_s m \quad \text{heat } [Q] = J, \text{ specific latent heat of fusion } [L_s] = J/kg, \text{ mass } [m] = kg$$

$$Q = L_f m \quad \text{heat } [Q] = J, \text{ specific latent heat of vaporization } [L_f] = J/kg, \text{ mass } [m] = kg$$

$$p = \rho gh \quad \text{pressure } [p] = Pa, \text{ density } [\rho] = \frac{kg}{m^3}, \text{ gravitational acceleration } [g] = \frac{m}{s^2}, \text{ height } [h] = m$$

$$p = p_1 + p_2 \quad \text{total pressure } [p] = Pa, \text{ partial pressure 1 } [p_1] = Pa, \text{ partial pressure 2 } [p_2] = Pa$$

$$p = p_{air} + p_{vapor} \quad \text{total pressure } [p] = Pa, \text{ air pressure } [p_{air}] = Pa, \text{ vapor pressure } [p_{vapor}] = Pa$$

$$T = (t + 273)K \quad \text{absolute temperature } [T] = K, \text{ temperature } [t] = ^\circ C$$

$$pV = nRT \quad \text{pressure } [p] = Pa, \text{ volume } [V] = m^3, \text{ amount of substance } [n] = mol, \text{ gas constant } [R] = \frac{Pa \cdot m^3}{mol \cdot K}, \text{ absol}$$

$$\frac{p_i V_i}{T_i} = \frac{p_f V_f}{T_f} \quad \text{initial pressure } [p_i] = Pa, \text{ initial volume } [V_i] = m^3, \text{ initial temperature } [T_i] = K, \text{ final pressure } [p_f] =$$

$$A = -p \Delta V \quad \text{work done } [A] = J, \text{ constant pressure } [p] = Pa, \text{ change in volume } [\Delta V] = m^3$$

$$I = \frac{Q}{t} \quad \text{current } [I] = A, \text{ charge } [Q] = C, \text{ time } [t] = s$$

$$U = \frac{E}{Q} \quad \text{voltage } [U] = V, \text{ energy } [E] = J, \text{ charge } [Q] = C$$

$$|Q| = Ne \quad \text{total charge } [|Q|] = C, \text{ number of elementary charges } [N], \text{ elementary charge } [e] = C$$

$$U = RI \quad \text{voltage } [U] = V, \text{ resistance } [R] = \Omega, \text{ current } [I] = A$$

$$P = UI \quad \text{power } [P] = W, \text{ voltage } [U] = V, \text{ current } [I] = A$$

$$P = RI^2 \quad \text{power } [P] = W, \text{ resistance } [R] = \Omega, \text{ current } [I] = A$$

$$\begin{aligned}
P &= \frac{U^2}{R} & \text{power } [P] = W, \text{ voltage } [U] = V, \text{ resistance } [R] = \Omega \\
R &= R_1 + R_2 & \text{total resistance } [R] = \Omega, \text{ resistance 1 } [R_1] = \Omega, \text{ resistance 2 } [R_2] = \Omega \\
\frac{1}{R} &= \frac{1}{R_1} + \frac{1}{R_2} & \text{total resistance } [R] = \Omega, \text{ resistance 1 } [R_1] = \Omega, \text{ resistance 2 } [R_2] = \Omega \\
R &= \rho \frac{l}{A} & \text{resistance } [R] = \Omega, \text{ resistivity } [\rho] = \Omega \cdot m, \text{ length } [l] = m, \text{ cross-sectional area } [A] = m^2 \\
\rho_t &= \rho_0(1 + \alpha_0 t) & \text{resistivity at temperature } t [\rho_t] = \Omega \cdot m, \text{ resistivity at } 0^\circ C [\rho_0] = \Omega \cdot m, \text{ temperature coefficient } [\alpha_0] = \text{ }^\circ C^{-1} \\
R_t &= R_0(1 + \alpha_0 t) & \text{resistance at temperature } t [R_t] = \Omega, \text{ resistance at } 0^\circ C [R_0] = \Omega, \text{ temperature coefficient } [\alpha_0] = \text{ }^\circ C^{-1} \\
U_p &= -R_i I + U_0 & \text{terminal voltage } [U_p] = V, \text{ internal resistance } [R_i] = \Omega, \text{ current } [I] = A, \text{ open-circuit voltage } [U_0] = V \\
f &= \frac{1}{T} & \text{frequency } [f] = Hz, \text{ period } [T] = s \\
v &= \lambda f & \text{wave speed } [v] = \frac{m}{s}, \text{ wavelength } [\lambda] = m, \text{ frequency } [f] = Hz \\
n &= \frac{c}{v} & \text{refractive index } [n], \text{ speed of light in vacuum } [c] = \frac{m}{s}, \text{ speed of light in material } [v] = \frac{m}{s} \\
\frac{\sin i}{\sin b} &= \frac{\lambda_1}{\lambda_2} = \frac{v_1}{v_2} = \frac{n_2}{n_1} & \text{angle of incidence } [i], \text{ angle of refraction } [b], \text{ wavelength in material 1 } [\lambda_1] = m, \text{ wavelength in material 2 } [\lambda_2] = m \\
i &= u & \text{angle of incidence } [i], \text{ angle of reflection } [u] \\
\sin i_c &= \frac{n_2}{n_1} & \text{critical angle } [i_c], \text{ refractive index of material 2 } [n_2], \text{ refractive index of material 1 } [n_1] \\
n\lambda &= d \sin \varphi_n & \text{order of maximum } [n], \text{ wavelength } [\lambda] = m, \text{ grating constant } [d] = m, \text{ diffraction angle } [\varphi_n] \\
E_{\text{photon}} &= hf & \text{photon energy } [E_{\text{photon}}] = J, \text{ Planck's constant } [h] = J \cdot s, \text{ frequency } [f] = Hz \\
E_{\text{photon}} &= E_n - E_m & \text{photon energy } [E_{\text{photon}}] = J, \text{ energy level n } [E_n] = J, \text{ energy level m } [E_m] = J \\
v_{\text{sound}} &= 331 \sqrt{\frac{T}{273K}} \frac{m}{s} & \text{speed of sound in air } [v_{\text{sound}}] = \frac{m}{s}, \text{ air temperature } [T] = K \\
\mu &= \frac{m_{\text{string}}}{L} & \text{linear mass density } [\mu] = \frac{kg}{m}, \text{ string mass } [m_{\text{string}}] = kg, \text{ string length } [L] = m \\
v_{\text{string}} &= \sqrt{\frac{F}{\mu}} & \text{string wave speed } [v_{\text{string}}] = \frac{m}{s}, \text{ string tension } [F] = N, \text{ linear mass density } [\mu] = \frac{kg}{m} \\
v &= \frac{\Delta s}{\Delta t} & \text{velocity } [v] = \frac{m}{s}, \text{ change in displacement } [\Delta s] = m, \text{ change in time } [\Delta t] = s \\
s &= vt + s_0 & \text{displacement } [s] = m, \text{ velocity } [v] = \frac{m}{s}, \text{ time } [t] = s, \text{ initial displacement } [s_0] = m \\
a &= \frac{\Delta v}{\Delta t} & \text{acceleration } [a] = \frac{m}{s^2}, \text{ change in velocity } [\Delta v] = \frac{m}{s}, \text{ change in time } [\Delta t] = s \\
v &= at + v_0 & \text{velocity } [v] = \frac{m}{s}, \text{ acceleration } [a] = \frac{m}{s^2}, \text{ time } [t] = s, \text{ initial velocity } [v_0] = \frac{m}{s} \\
s &= \frac{1}{2} at^2 + v_0 t + s_0 & \text{displacement } [s] = m, \text{ acceleration } [a] = \frac{m}{s^2}, \text{ time } [t] = s, \text{ initial velocity } [v_0] = \frac{m}{s}, \text{ initial displacement } [s_0] = m \\
v^2 &= v_0^2 + 2a(s - s_0) & \text{final velocity } [v] = \frac{m}{s}, \text{ initial velocity } [v_0] = \frac{m}{s}, \text{ acceleration } [a] = \frac{m}{s^2}, \text{ final displacement } [s] = m \\
F_g &= mg & \text{gravitational force } [F_g] = N, \text{ mass } [m] = kg, \text{ gravitational acceleration } [g] = \frac{m}{s^2} \\
F_{\text{spring}} &= -kx & \text{spring force } [F_{\text{spring}}] = N, \text{ spring constant } [k] = \frac{N}{m}, \text{ displacement from equilibrium } [x] = m \\
F_{\text{buoyancy}} &= \rho_{\text{liquid}} V g & \text{buoyant force } [F_{\text{buoyancy}}] = N, \text{ density of liquid/gas } [\rho_{\text{liquid}}] = \frac{kg}{m^3}, \text{ volume of object } [V] = m^3 \\
F_f &= \mu F_N & \text{friction force } [F_f] = N, \text{ coefficient of friction } [\mu], \text{ normal force } [F_N] = N \\
F_{\text{drag}} &= \frac{1}{2} c_W \rho A v^2 & \text{air resistance } [F_{\text{drag}}] = N, \text{ drag coefficient } [c_W], \text{ air density } [\rho] = \frac{kg}{m^3}, \text{ cross-sectional area } [A] = m^2
\end{aligned}$$

$$F_{net} = ma \quad \text{net force } [F_{net}] = N, \text{ mass } [m] = kg, \text{ acceleration } [a] = \frac{m}{s^2}$$

$$\Delta E = 0 \quad \text{change in energy } [\Delta E] = J$$

$$W = Fs \cos(\phi) \quad \text{work } [W] = J, \text{ force } [F] = N, \text{ displacement } [s] = m, \text{ angle between force and displacement } [\phi]$$

$$W_{spring} = -\frac{1}{2}kx^2 \quad \text{work done by spring } [W_{spring}] = J, \text{ spring constant } [k] = \frac{N}{m}, \text{ displacement from equilibrium } [x] = m$$

$$P = \frac{W}{t} = Fv \quad \text{power } [P] = W, \text{ work } [W] = J, \text{ time } [t] = s, \text{ force } [F] = N, \text{ velocity } [v] = \frac{m}{s}$$

$$E_{kin} = \frac{1}{2}mv^2 \quad \text{kinetic energy } [E_{kin}] = J, \text{ mass } [m] = kg, \text{ velocity } [v] = \frac{m}{s}$$

$$E_{pot} = mgh \quad \text{potential energy } [E_{pot}] = J, \text{ mass } [m] = kg, \text{ gravitational acceleration } [g] = \frac{m}{s^2}, \text{ height } [h] = m$$

$$E_{mech} = E_{kin} + E_{pot} \quad \text{mechanical energy } [E_{mech}] = J, \text{ kinetic energy } [E_{kin}] = J, \text{ potential energy } [E_{pot}] = J$$

$$I = I_0 \left(\frac{1}{2}\right)^{\frac{x}{x_{1/2}}} \quad \text{intensity } [I], \text{ initial intensity } [I_0], \text{ material thickness } [x] = m, \text{ half-thickness } [x_{1/2}] = m$$

$$N = N_0 \left(\frac{1}{2}\right)^{\frac{t}{T_{1/2}}} \quad \text{number of nuclei } [N], \text{ initial number of nuclei } [N_0], \text{ time } [t] = s, \text{ half-life } [T_{1/2}] = s$$

$$A = A_0 \left(\frac{1}{2}\right)^{\frac{t}{T_{1/2}}} \quad \text{activity } [A] = Bq, \text{ initial activity } [A_0] = Bq, \text{ time } [t] = s, \text{ half-life } [T_{1/2}] = s$$

$$A = kN \quad \text{activity } [A] = Bq, \text{ decay constant } [k] = s^{-1}, \text{ number of nuclei } [N]$$

$$k = \frac{\ln(2)}{T_{1/2}} \quad \text{decay constant } [k] = s^{-1}, \text{ half-life } [T_{1/2}] = s$$

$$E = mc^2 \quad \text{energy } [E] = J, \text{ rest mass } [m] = kg, \text{ speed of light in vacuum } [c] = \frac{m}{s}$$

$$E_b = M_d c^2 \quad \text{binding energy } [E_b] = J, \text{ mass defect } [M_d] = kg, \text{ speed of light in vacuum } [c] = \frac{m}{s}$$

$$M_d = Zm_p + Nm_n - M_k \quad \text{mass defect } [M_d] = kg, \text{ number of protons } [Z], \text{ proton mass } [m_p] = kg, \text{ number of neutrons } [N], \text{ mass of nucleus } [M_k]$$

$$M_{nucleus} = M_{atom} - Zm_e \quad \text{nuclear mass } [M_{nucleus}] = kg, \text{ atomic mass } [M_{atom}] = kg, \text{ number of protons } [Z], \text{ electron mass } [m_e]$$

$$\Delta S = nR \ln\left(\frac{T_2}{T_1}\right) + nC_{im} \ln\left(\frac{T_2}{T_1}\right) \quad \text{change in entropy for reversible isobaric process } [\Delta S] = J/K, \text{ amount of substance } [n] = mol$$

$$W = -p\Delta V = -nR\Delta T \quad \text{work done for reversible isobaric process } [W] = J, \text{ pressure } [p] = Pa, \text{ change in volume } [\Delta V]$$

$$\Delta U = nC_{vm}\Delta T \quad \text{change in internal energy for isobaric process } [\Delta U] = J, \text{ amount of substance } [n] = mol, \text{ molar heat capacity } [C_{vm}]$$

$$\Delta H = nC_{pm}\Delta T \quad \text{change in enthalpy for isobaric process } [\Delta H] = J, \text{ amount of substance } [n] = mol, \text{ molar heat capacity } [C_{pm}]$$

$$\Delta S = nC_{vm}\Delta T \quad \text{change in entropy for reversible isochoric process } [\Delta S] = J/K, \text{ amount of substance } [n] = mol, \text{ molar heat capacity } [C_{vm}]$$

$$W = 0 \quad \text{work done for reversible isochoric process } [W] = J$$

$$\Delta U = nC_{vm}\Delta T \quad \text{change in internal energy for isochoric process } [\Delta U] = J, \text{ amount of substance } [n] = \text{mol}, \text{ molar heat capacity } [C_{vm}] = J/mol \cdot K$$

$$\Delta H = nC_{pm}\Delta T \quad \text{change in enthalpy for isochoric process } [\Delta H] = J, \text{ amount of substance } [n] = \text{mol}, \text{ molar heat capacity } [C_{pm}] = J/mol \cdot K$$

$$\Delta S = nR \ln\left(\frac{V_2}{V_1}\right) \quad \text{change in entropy for reversible isothermal process } [\Delta S] = J/K, \text{ amount of substance } [n] = \text{mol}, \text{ gas constant } [R] = J/mol \cdot K$$

$$W = -nRT \ln\left(\frac{V_2}{V_1}\right) \quad \text{work done for reversible isothermal process } [W] = J, \text{ amount of substance } [n] = \text{mol}, \text{ gas constant } [R] = J/mol \cdot K$$

$$\Delta U = 0 \quad \text{change in internal energy for isothermal process } [\Delta U] = J$$

$$\Delta H = 0 \quad \text{change in enthalpy for isothermal process } [\Delta H] = J$$

$$\Delta S = nR \ln\left(\frac{V_2}{V_1}\right) + nC_{vm} \ln\left(\frac{T_2}{T_1}\right) \quad \text{change in entropy for reversible adiabatic process } [\Delta S] = J/K, \text{ amount of substance } [n] = \text{mol}, \text{ gas constant } [R] = J/mol \cdot K$$

$$W = nC_{vm}\Delta T \quad \text{work done for reversible adiabatic process } [W] = J, \text{ amount of substance } [n] = \text{mol}, \text{ molar heat capacity } [C_{vm}] = J/mol \cdot K$$

$$\Delta U = nC_{vm}\Delta T \quad \text{change in internal energy for adiabatic process } [\Delta U] = J, \text{ amount of substance } [n] = \text{mol}, \text{ molar heat capacity } [C_{vm}] = J/mol \cdot K$$

$$\Delta H = nC_{pm}\Delta T \quad \text{change in enthalpy for adiabatic process } [\Delta H] = J, \text{ amount of substance } [n] = \text{mol}, \text{ molar heat capacity } [C_{pm}] = J/mol \cdot K$$

$$C_{pm} = C_{vm} + R \quad \text{molar heat capacity at constant pressure } [C_{pm}] = \frac{J}{mol \cdot K}, \text{ molar heat capacity at constant volume } [C_{vm}] = J/mol \cdot K$$

$$\left(\frac{\partial T}{\partial V}\right)_S = \left(\frac{\partial p}{\partial S}\right)_V \quad \text{Maxwell relation for U}$$

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p \quad \text{Maxwell relation for H}$$

$$\left(\frac{\partial p}{\partial T}\right)_V = \left(\frac{\partial S}{\partial V}\right)_T \quad \text{Maxwell relation for A}$$

$$\left(\frac{\partial V}{\partial T}\right)_p = -\left(\frac{\partial S}{\partial p}\right)_T \quad \text{Maxwell relation for G}$$

$$\Delta S = nR \ln\left(\frac{V_2}{V_1}\right) \quad \text{change in entropy } [\Delta S] = J/K, \text{ amount of substance } [n] = \text{mol}, \text{ gas constant } [R] = \frac{J}{mol \cdot K}, \text{ volume } [V] = m^3$$

$$V_m = \frac{M}{\rho} \quad \text{molar volume } [V_m] = m^3/mol, \text{ molar mass } [M] = kg/mol, \text{ density } [\rho] = kg/m^3$$

$$p = \frac{nRT}{V - nb} - a\left(\frac{n}{V}\right)^2 \quad \text{Van der Waals equation of state: pressure } [p] = Pa, \text{ amount of substance } [n] = \text{mol}, \text{ gas constant } [R] = J/mol \cdot K$$

$$p = \frac{RT}{V_m - b} - \frac{a}{V_m^2} \quad \text{Van der Waals equation of state (molar form): pressure } [p] = Pa, \text{ gas constant } [R] = \frac{J}{mol \cdot K}, \text{ temperature } [T] = K$$

$$a = 3p_c V_c^2 \quad \text{Van der Waals constant a } [a] = Pa \cdot m^6/mol^2, \text{ critical pressure } [p_c] = Pa, \text{ critical volume } [V_c] = m^3$$

$$b = \frac{1}{3}V_c \quad \text{Van der Waals constant b } [b] = m^3/mol, \text{ critical volume } [V_c] = m^3$$

$$\begin{aligned}
p_r &= \frac{p}{p_c} && \text{reduced pressure } [p_r], \text{ pressure } [p] = Pa, \text{ critical pressure } [p_c] = Pa \\
T_r &= \frac{T}{T_c} && \text{reduced temperature } [T_r], \text{ temperature } [T] = K, \text{ critical temperature } [T_c] = K \\
V_r &= \frac{V}{V_c} && \text{reduced volume } [V_r], \text{ volume } [V] = m^3, \text{ critical volume } [V_c] = m^3 \\
p_r &= \frac{8T_r}{3V_r - 1} - \frac{3}{V_r^2} && \text{Van der Waals equation with reduced variables: reduced pressure } [p_r], \text{ reduced temperature } [T_r] \\
\pi_T &= a\left(\frac{n}{V}\right)^2 && \text{internal pressure } [\pi_T] = Pa, \text{ Van der Waals constant } a [a] = Pa \cdot m^6/mol^2, \text{ amount of substance } [n] = mol \\
\Delta S &= \int_i^f \frac{dq_{rev}}{T} && \text{definition of entropy: change in entropy } [\Delta S] = J/K, \text{ reversible heat transfer } [dq_{rev}] = J, \text{ temperature } [T] = K \\
dS &\geq \frac{dq}{T} && \text{Clausius inequality: change in entropy } [dS] = J/K, \text{ heat transfer } [dq] = J, \text{ temperature } [T] = K \\
\Delta S &= nR \ln\left(\frac{V_2}{V_1}\right) + nC_{V,m} \ln\left(\frac{T_2}{T_1}\right) && \text{change in entropy } [\Delta S] = J/K, \text{ amount of substance } [n] = mol, \text{ gas constant } [R] = J/mol \cdot K \\
\Delta S &= nR \ln\left(\frac{V_2}{V_1}\right) = -nR \ln\left(\frac{p_2}{p_1}\right) && \text{change in entropy for isothermal ideal gas } [\Delta S] = J/K, \text{ amount of substance } [n] = mol \\
\Delta S_{sur} &= \frac{q_{sur}}{T_{sur}} && \text{change in entropy of surroundings } [\Delta S_{sur}] = J/K, \text{ heat of surroundings } [q_{sur}] = J, \text{ temperature of surroundings } [T_{sur}] = K \\
\Delta S_{sur} &= \frac{\Delta H_{sur}}{T_{sur}} && \text{change in entropy of surroundings } [\Delta S_{sur}] = J/K, \text{ change in enthalpy of surroundings } [\Delta H_{sur}] = J \\
\Delta S_{sur} &= 0 && \text{change in entropy of surroundings for adiabatic process } [\Delta S_{sur}] = J/K \\
\Delta S_{sur} &= 0 && \text{change in entropy of surroundings for reversible process } [\Delta S_{sur}] = J/K \\
a &= \frac{a_0}{Z} && \text{constant } a [a], \text{ Bohr radius } [a_0] = m, \text{ atomic number } [Z] \\
p &= \frac{h}{\lambda} && \text{De Broglie relation: linear momentum } [p] = kg \cdot m/s, \text{ Planck's constant } [h] = J \cdot s, \text{ wavelength } [\lambda] = m \\
p &= \frac{h}{\lambda} = \frac{nh}{2L} && \text{linear momentum for particle-in-a-box } [p] = kg \cdot m/s, \text{ Planck's constant } [h] = J \cdot s, \text{ wavelength } [\lambda] = m \\
\Psi_n(x) &= \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) && \text{wavefunction for particle-in-a-box } [\Psi_n(x)], \text{ box length } [L] = m, \text{ quantum number } [n], \text{ position } [x] = m \\
E_n &= \frac{n^2 h^2}{8mL^2} && \text{energy for particle-in-a-box } [E_n] = J, \text{ quantum number } [n], \text{ Planck's constant } [h] = J \cdot s, \text{ mass } [m] = kg \\
E_{n+1} - E_n &= (2n+1) \frac{h^2}{8mL^2} && \text{energy difference for particle-in-a-box } [E_{n+1} - E_n] = J, \text{ quantum number } [n], \text{ Planck's constant } [h] = J \cdot s \\
|\Psi|^2 &= \frac{2}{L} \sin^2\left(\frac{n\pi x}{L}\right) && \text{probability density for particle-in-a-box } [|\Psi|^2], \text{ box length } [L] = m, \text{ quantum number } [n], \text{ position } [x] = m \\
\Psi \Psi_n d\tau &= 0 && \text{orthogonality condition for wavefunctions} \\
\tilde{\nu} &= \frac{v}{c} = R\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right) && \text{Rydberg formula: wavenumber } [\tilde{\nu}] = m^{-1}, \text{ frequency } [v] = Hz, \text{ speed of light } [c] = m/s, \text{ Rydberg constant } [R] = m^{-1} \\
\tilde{\nu} &= Z^2 R_\alpha \left| \frac{1}{n_1^2} - \frac{1}{n_2^2} \right| && \text{emission of light from hydrogenic atoms: wavenumber } [\tilde{\nu}] = m^{-1}, \text{ atomic number } [Z], \text{ Rydberg constant } [R_\alpha] = m^{-1}
\end{aligned}$$

$F = ma$ Newton's Second Law: force $[F] = N$, mass $[m] = kg$, acceleration $[a] = \frac{m}{s^2}$

$E = h\nu$ Planck-Einstein relation: energy $[E] = J$, Planck's constant $[h] = J \cdot s$, frequency $[\nu] = Hz$

$G = H - TS$ Gibbs Free Energy: Gibbs free energy $[G] = J$, enthalpy $[H] = J$, temperature $[T] = K$, entropy $[S] = J/K$

$\Delta G = \Delta H - T\Delta S$ Change in Gibbs Free Energy: change in Gibbs free energy $[\Delta G] = J$, change in enthalpy $[\Delta H] = J$, temperature $[T] = K$

$\Delta G^\circ = -RT \ln K$ Gibbs Free Energy and Equilibrium Constant: standard change in Gibbs free energy $[\Delta G^\circ] = J$, gas constant $[R] = J/mol \cdot K$, equilibrium constant $[K]$

$\Delta G = \Delta G^\circ + RT \ln Q$ Gibbs Free Energy and Reaction Quotient: change in Gibbs free energy $[\Delta G] = J$, standard change in Gibbs free energy $[\Delta G^\circ] = J$, gas constant $[R] = J/mol \cdot K$, temperature $[T] = K$, reaction quotient $[Q]$

$E = E^\circ - \frac{RT}{nF} \ln Q$ Nernst Equation: cell potential $[E] = V$, standard cell potential $[E^\circ] = V$, gas constant $[R] = J/mol \cdot K$, temperature $[T] = K$, number of electrons transferred $[n]$, Faraday constant $[F] = C/mol$

$k = Ae^{-\frac{E_a}{RT}}$ Arrhenius Equation: rate constant $[k]$, pre-exponential factor $[A]$, activation energy $[E_a] = J/mol$, gas constant $[R] = J/mol \cdot K$, temperature $[T] = K$

$PV = nRT$ Ideal Gas Law: pressure $[P] = Pa$, volume $[V] = m^3$, amount of substance $[n] = mol$, gas constant $[R] = J/mol \cdot K$, temperature $[T] = K$

$\lambda_{max} = \frac{b}{T}$ Wien's Displacement Law: wavelength of maximum emission $[\lambda_{max}] = m$, Wien's displacement constant $[b] = m \cdot K$

$\rho_{rad} = \frac{8\pi h}{c^3 \lambda^5} \frac{1}{e^{\frac{hc}{\lambda kT}} - 1}$ Planck's Law (spectral radiance): spectral energy density $[\rho_{rad}] = J/m^3 \cdot m$, Planck's constant $[h] = J \cdot s$, speed of light $[c] = m/s$, wavelength $[\lambda] = m$, Boltzmann constant $[k] = J/K$, temperature $[T] = K$

$F = G \frac{m_1 m_2}{r^2}$ Newton's Law of Universal Gravitation: gravitational force $[F] = N$, gravitational constant $[G] = \frac{N \cdot m^2}{kg^2}$, masses $[m_1] = kg$, $[m_2] = kg$, distance $[r] = m$

$\tau = rF \sin \theta$ Torque: torque $[\tau] = N \cdot m$, lever arm $[r] = m$, force $[F] = N$, angle between r and F $[\theta]$

$L = I\omega$ Angular Momentum: angular momentum $[L] = \frac{kg \cdot m^2}{s}$, moment of inertia $[I] = kg \cdot m^2$, angular velocity $[\omega] = \frac{1}{s}$

$\frac{d[A]}{dt} = -k[A]$ First-Order Rate Law (differential form): rate of change of concentration of A $[\frac{d[A]}{dt}] = \frac{mol}{L \cdot s}$, rate constant $[k]$

$[A]_t = [A]_0 e^{-kt}$ First-Order Rate Law (integrated form): concentration of A at time t $[[A]_t] = \frac{mol}{L}$, initial concentration $[A]_0 = \frac{mol}{L}$, rate constant $[k] = s^{-1}$

$t_{1/2} = \frac{\ln 2}{k}$ Half-life for First-Order Reaction: half-life $[t_{1/2}] = s$, rate constant $[k] = s^{-1}$

$E_{kin,rot} = \frac{1}{2}I\omega^2$ Rotational Kinetic Energy: rotational kinetic energy $[E_{kin,rot}] = J$, moment of inertia $[I] = kg \cdot m^2$

$\tau = I\alpha$ Newton's Second Law for Rotation: torque $[\tau] = N \cdot m$, moment of inertia $[I] = kg \cdot m^2$, angular acceleration $[\alpha] = rad/s^2$

$F = k_e \frac{|q_1 q_2|}{r^2}$ Coulomb's Law: electric force $[F] = N$, Coulomb's constant $[k_e] = \frac{N \cdot m^2}{C^2}$, charge 1 $[q_1] = C$, charge 2 $[q_2] = C$, distance $[r] = m$

$E = \frac{F}{q_0}$ Electric Field: electric field $[E] = \frac{N}{C}$, electric force $[F] = N$, test charge $[q_0] = C$

$U = k_e \frac{q_1 q_2}{r}$ Electric Potential Energy: electric potential energy $[U] = J$, Coulomb's constant $[k_e] = \frac{N \cdot m^2}{C^2}$, charge 1 $[q_1] = C$, charge 2 $[q_2] = C$, distance $[r] = m$

$V = \frac{U}{q_0} = k_e \frac{q}{r}$ Electric Potential: electric potential $[V] = V$, electric potential energy $[U] = J$, test charge $[q_0] = C$, distance $[r] = m$

$C = \frac{Q}{V}$ Capacitance: capacitance $[C] = F$, charge $[Q] = C$, voltage $[V] = V$

$E_{cap} = \frac{1}{2}CV^2 = \frac{1}{2} \frac{Q^2}{C} = \frac{1}{2}QV$ Energy Stored in Capacitor: energy $[E_{cap}] = J$, capacitance $[C] = F$, voltage $[V] = V$, charge $[Q] = C$

$F = ILB \sin \theta$ Force on a Current-Carrying Wire in Magnetic Field: force $[F] = N$, current $[I] = A$, length of wire $[L] = m$, magnetic field strength $[B] = T$, angle $[\theta] = \text{degrees}$

$F = qvB \sin \theta$ Force on a Moving Charge in Magnetic Field (Lorentz Force): force $[F] = N$, charge $[q] = C$, velocity $[v] = m/s$, magnetic field strength $[B] = T$, angle $[\theta] = \text{degrees}$

$\Phi_B = BA \cos \theta$ Magnetic Flux: magnetic flux $[\Phi_B] = Wb$, magnetic field strength $[B] = T$, area $[A] = m^2$, angle between $[B]$ and $[A]$ $[\theta] = \text{degrees}$

$\epsilon = -N \frac{\Delta \Phi_B}{\Delta t}$ Faraday's Law of Induction: induced EMF $[\epsilon] = V$, number of turns $[N]$, change in magnetic flux $[\Delta \Phi_B] = Wb$, time $[\Delta t] = s$

$X_L = \omega L$ Inductive Reactance: inductive reactance $[X_L] = \Omega$, angular frequency $[\omega] = \frac{rad}{s}$, inductance $[L] = H$

$X_C = \frac{1}{\omega C}$ Capacitive Reactance: capacitive reactance $[X_C] = \Omega$, angular frequency $[\omega] = \frac{rad}{s}$, capacitance $[C] = F$

$Z = \sqrt{R^2 + (X_L - X_C)^2}$ Impedance (RLC Series Circuit): impedance $[Z] = \Omega$, resistance $[R] = \Omega$, inductive reactance $[X_L] = \Omega$, capacitive reactance $[X_C] = \Omega$

$E = mc^2$ Mass-Energy Equivalence: energy $[E] = J$, mass $[m] = kg$, speed of light $[c] = \frac{m}{s}$

$\Delta H_{rxn}^\circ = \sum n \Delta H_f^\circ(products) - \sum m \Delta H_f^\circ(reactants)$ Standard Enthalpy of Reaction from Enthalpies of Formation

$\Delta S_{rxn}^\circ = \sum n S^\circ(products) - \sum m S^\circ(reactants)$ Standard Entropy of Reaction from Standard Molar Entropies: standard molar entropy $[S^\circ] = J/K$

$\Delta G_{rxn}^\circ = \sum n \Delta G_f^\circ(products) - \sum m \Delta G_f^\circ(reactants)$ Standard Gibbs Free Energy of Reaction from Standard Gibbs Free Energies of Formation

$pH = -\log_{10}[H^+]$ pH Definition: pH, hydrogen ion concentration $[[H^+]] = M$

$pOH = -\log_{10}[OH^-]$ pOH Definition: pOH, hydroxide ion concentration $[[OH^-]] = M$

$pH + pOH = 14$ pH and pOH relationship

$K_a = \frac{[H^+][A^-]}{[HA]}$ Acid Dissociation Constant: acid dissociation constant $[K_a]$, hydrogen ion concentration $[[H^+]] = M$, conjugate base concentration $[[A^-]] = M$, acid concentration $[[HA]] = M$

$K_b = \frac{[BH^+][OH^-]}{[B]}$ Base Dissociation Constant: base dissociation constant $[K_b]$, conjugate acid concentration $[[BH^+]] = M$, base concentration $[[B]] = M$, hydroxide ion concentration $[[OH^-]] = M$

$E = hf - \Phi$ Photoelectric Effect (Kinetic Energy of Emitted Electron): kinetic energy of electron $[E] = J$, Planck's constant $[h] = J \cdot s$, frequency $[f] = Hz$, work function $[\Phi] = J$

$\Delta x \Delta p \geq \frac{\hbar}{2}$ Heisenberg Uncertainty Principle: uncertainty in position $[\Delta x] = m$, uncertainty in momentum $[\Delta p] = kg \cdot m/s$

$c = \lambda f$ Speed of Light: speed of light $[c] = \frac{m}{s}$, wavelength $[\lambda] = m$, frequency $[f] = Hz$

$F_{centripetal} = \frac{mv^2}{r} = m\omega^2 r$ Centripetal Force: centripetal force $[F_{centripetal}] = N$, mass $[m] = kg$, tangential velocity $[v] = \frac{m}{s}$

$KE_{avg} = \frac{3}{2}k_B T$ Average Translational Kinetic Energy of Gas: average kinetic energy $[KE_{avg}] = J$, Boltzmann constant $[k_B] = \frac{J}{K}$

$v_{rms} = \sqrt{\frac{3k_B T}{m}} = \sqrt{\frac{3RT}{M}}$ Root-Mean-Square Speed of Gas: root-mean-square speed $[v_{rms}] = \frac{m}{s}$, Boltzmann constant $[k_B] = \frac{J}{K}$, molar mass $[M] = \frac{kg}{mol}$

$\mu = \frac{m}{n}$ Molar Mass (alternative): molar mass $[\mu] = \frac{kg}{mol}$, mass $[m] = kg$, amount of substance $[n] = mol$

$\Delta T_f = K_f m$ Freezing Point Depression: change in freezing point $[\Delta T_f] = K$, cryoscopic constant $[K_f] = \frac{K \cdot kg}{mol}$, molality $[m] = \frac{mol}{kg}$

$\Delta T_b = K_b m$ Boiling Point Elevation: change in boiling point $[\Delta T_b] = K$, ebullioscopic constant $[K_b] = \frac{K \cdot kg}{mol}$, molality $[m] = \frac{mol}{kg}$

$\Pi = iMRT$ Osmotic Pressure: osmotic pressure $[\Pi] = Pa$, van't Hoff factor $[i]$, molarity $[M] = \frac{mol}{L}$, gas constant $[R] = \frac{J}{mol \cdot K}$, temperature $[T] = K$

$E = \frac{hc}{\lambda}$ Photon Energy (alternative): energy $[E] = J$, Planck's constant $[h] = J \cdot s$, speed of light $[c] = \frac{m}{s}$, wavelength $[\lambda] = m$

$\lambda = \frac{h}{p} = \frac{h}{mv}$ De Broglie Wavelength (alternative): wavelength $[\lambda] = m$, Planck's constant $[h] = J \cdot s$, momentum $[p] = \frac{kg \cdot m}{s}$

$\hat{H}\Psi = E\Psi$ Time-Independent Schrodinger Equation: Hamiltonian operator $[\hat{H}]$, wavefunction $[\Psi]$, energy $[E] = J$

$\Delta E_{vib} = h\nu_{osc} = h\frac{1}{2\pi}\sqrt{\frac{k}{\mu}}$ Vibrational Energy (Quantum Harmonic Oscillator): vibrational energy change $[\Delta E_{vib}] = J$, reduced mass $[\mu] = \frac{kg}{mol}$, spring constant $[k] = \frac{N}{m}$

$\Delta E_{rot} = BJ(J+1)$ Rotational Energy (Rigid Rotor): rotational energy change $[\Delta E_{rot}] = J$, rotational constant $[B] = \frac{J}{mol}$, moment of inertia $[I] = \frac{kg \cdot m^2}{mol}$

$$a = r\alpha \quad \text{Tangential Acceleration: tangential acceleration } [a] = \frac{m}{s^2}, \text{ radius } [r] = m, \text{ angular acceleration } [\alpha] = \frac{rad}{s^2}$$

$$v = r\omega \quad \text{Tangential Velocity: tangential velocity } [v] = \frac{m}{s}, \text{ radius } [r] = m, \text{ angular velocity } [\omega] = \frac{rad}{s}$$

$$\omega = \omega_0 + \alpha t \quad \text{Angular Velocity with Constant Angular Acceleration: final angular velocity } [\omega] = \frac{rad}{s}, \text{ initial angular velocity } [\omega_0] = \frac{rad}{s}$$

$$\theta = \omega_0 t + \frac{1}{2}\alpha t^2 \quad \text{Angular Displacement with Constant Angular Acceleration: angular displacement } [\theta] = rad, \text{ initial angular displacement } [\theta_0] = rad$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta \quad \text{Angular Velocity Squared with Constant Angular Acceleration: final angular velocity } [\omega] = \frac{rad}{s}, \text{ initial angular velocity } [\omega_0] = \frac{rad}{s}$$

$$I = \sum m_i r_i^2 \quad \text{Moment of Inertia (Discrete Masses): moment of inertia } [I] = kg \cdot m^2, \text{ mass of particle } i [m_i] = kg, \text{ distance from axis } [r_i] = m$$

$$KE_{rot} = \frac{1}{2}I\omega^2 \quad \text{Rotational Kinetic Energy: rotational kinetic energy } [KE_{rot}] = J, \text{ moment of inertia } [I] = kg \cdot m^2, \text{ angular velocity } [\omega] = \frac{rad}{s}$$

$$L = I\omega \quad \text{Angular Momentum: angular momentum } [L] = \frac{kg \cdot m^2}{s}, \text{ moment of inertia } [I] = kg \cdot m^2, \text{ angular velocity } [\omega] = \frac{rad}{s}$$

$$\tau = \frac{\Delta L}{\Delta t} \quad \text{Torque as Rate of Change of Angular Momentum: torque } [\tau] = N \cdot m, \text{ change in angular momentum } [\Delta L] = kg \cdot m^2/s$$

$$F_{grav} = -G \frac{m_1 m_2}{r^2} \hat{r} \quad \text{Newton's Law of Universal Gravitation (Vector Form): gravitational force } [F_{grav}] = N, \text{ gravitational constant } [G] = \frac{N \cdot m^2}{kg^2}$$

$$U_{grav} = -G \frac{m_1 m_2}{r} \quad \text{Gravitational Potential Energy: gravitational potential energy } [U_{grav}] = J, \text{ gravitational constant } [G] = \frac{N \cdot m^2}{kg^2}$$

$$v_{esc} = \sqrt{\frac{2GM}{R}} \quad \text{Escape Velocity: escape velocity } [v_{esc}] = \frac{m}{s}, \text{ gravitational constant } [G] = \frac{N \cdot m^2}{kg^2}, \text{ mass of planet/star } [M] = kg, \text{ radius } [R] = m$$

$$T^2 = \left(\frac{4\pi^2}{GM}\right)r^3 \quad \text{Kepler's Third Law (Period): orbital period } [T] = s, \text{ gravitational constant } [G] = \frac{N \cdot m^2}{kg^2}, \text{ mass of central body } [M] = kg, \text{ semi-major axis } [r] = m$$

$$P = \sigma AT^4 \quad \text{Stefan-Boltzmann Law: power radiated } [P] = W, \text{ Stefan-Boltzmann constant } [\sigma] = \frac{W}{m^2 \cdot K^4}, \text{ surface area } [A] = m^2, \text{ temperature } [T] = K$$

$$Q = \dot{Q}\Delta t = kA \frac{\Delta T}{L} \Delta t \quad \text{Heat Conduction (Rate): heat transferred } [Q] = J, \text{ heat transfer rate } [\dot{Q}] = W, \text{ thermal conductivity } [k] = \frac{W}{m \cdot K}, \text{ area } [A] = m^2, \text{ length } [L] = m, \text{ temperature difference } [\Delta T] = K$$

$$\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} \geq 0 \quad \text{Second Law of Thermodynamics (Entropy of Universe): change in entropy of universe } [\Delta S] = J/K$$

$$\mu_J = \left(\frac{\partial T}{\partial p}\right)_H \quad \text{Joule-Thomson Coefficient: Joule-Thomson coefficient } [\mu_J] = K/Pa, \text{ temperature } [T] = K, \text{ pressure } [p] = Pa$$

$$E_v = \frac{1}{2}\hbar\omega\left(v + \frac{1}{2}\right) \quad \text{Vibrational Energy Levels (Quantum Harmonic Oscillator): vibrational energy } [E_v] = J, \text{ reduced mass } [\mu] = kg, \text{ spring constant } [k] = N/m$$

$$E_J = BJ(J+1) \quad \text{Rotational Energy Levels (Rigid Rotor): rotational energy } [E_J] = J, \text{ rotational constant } [B] = J, \text{ quantum number } [J] = 0, 1, 2, \dots$$

$$P(v) = 4\pi\left(\frac{m}{2\pi k_B T}\right)^{3/2} v^2 e^{-\frac{mv^2}{2k_B T}} \quad \text{Maxwell-Boltzmann Speed Distribution: probability density function } [P(v)] = s/m^3$$

$$Z = \sum_i e^{-\frac{E_i}{k_B T}} \quad \text{Partition Function: partition function } [Z], \text{ energy of state } i [E_i] = J, \text{ Boltzmann constant } [k_B] = J/K$$

$$A = -k_B T \ln Z \quad \text{Helmholtz Free Energy from Partition Function: Helmholtz free energy } [A] = J, \text{ Boltzmann constant } [k_B] = J/K, \text{ temperature } [T] = K$$

$$S = k_B \ln W \quad \text{Boltzmann's Entropy Formula: entropy } [S] = J/K, \text{ Boltzmann constant } [k_B] = J/K, \text{ number of microstates } [W]$$

$$\ln K = -\frac{\Delta H^\circ}{RT} + \frac{\Delta S^\circ}{R} \quad \text{Van't Hoff Equation (Linearized): natural log of equilibrium constant } [\ln K], \text{ standard enthalpy change } [\Delta H^\circ] = J, \text{ standard entropy change } [\Delta S^\circ] = J/K, \text{ temperature } [T] = K$$

$$\ln\left(\frac{K_2}{K_1}\right) = -\frac{\Delta H^\circ}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right) \quad \text{Van't Hoff Equation (Two Temperatures): equilibrium constant at } T_2 [K_2], \text{ equilibrium constant at } T_1 [K_1], \text{ standard enthalpy change } [\Delta H^\circ] = J$$

$$\Delta_r G^\circ = -RT \ln K_p \quad \text{Gibbs Free Energy and Equilibrium Constant (Gas Phase): standard Gibbs free energy of reaction } [\Delta_r G^\circ] = J, \text{ gas constant } [R] = J/K, \text{ temperature } [T] = K$$

$$\frac{dP}{dT} = \frac{\Delta H_{vap}}{T\Delta V_{vap}} \quad \text{Clapeyron Equation: derivative of vapor pressure with respect to temperature } \left[\frac{dP}{dT}\right] = Pa/K, \text{ enthalpy of vaporization } [\Delta H_{vap}] = J, \text{ volume change } [\Delta V_{vap}] = m^3$$

$$\ln P = -\frac{\Delta H_{vap}}{RT} + C \quad \text{Clausius-Clapeyron Equation (Integrated): natural log of vapor pressure } [\ln P], \text{ enthalpy of vaporization } [\Delta H_{vap}] = J, \text{ gas constant } [R] = J/K, \text{ temperature } [T] = K$$

$$\Delta_{mix} G = nRT \sum x_i \ln x_i \quad \text{Gibbs Energy of Mixing (Ideal Solution): Gibbs energy of mixing } [\Delta_{mix} G] = J, \text{ total moles } [n] = mol, \text{ gas constant } [R] = J/K, \text{ temperature } [T] = K, \text{ mole fraction } [x_i]$$

$$\Delta_{mix} S = -nR \sum x_i \ln x_i \quad \text{Entropy of Mixing (Ideal Solution): entropy of mixing } [\Delta_{mix} S] = J/K, \text{ total moles } [n] = mol, \text{ gas constant } [R] = J/K, \text{ mole fraction } [x_i]$$

$$\Delta_{mix} H = 0 \quad \text{Enthalpy of Mixing (Ideal Solution): enthalpy of mixing } [\Delta_{mix} H] = J$$

$$a_i = \gamma_i x_i \quad \text{Activity of a Component: activity } [a_i], \text{ activity coefficient } [\gamma_i], \text{ mole fraction } [x_i]$$

$$E = -\vec{\nabla} V \quad \text{Electric Field as Negative Gradient of Potential: electric field } [E] = V/m, \text{ electric potential } [V] = V, \text{ del operator } [\vec{\nabla}]$$